

```
In [ ]: - list
        - tuple
        - set
        - dict
        - range
```

```
In [1]: l = []
        l
```

```
Out[1]: []
```

```
In [2]: len(l)
```

```
Out[2]: 0
```

```
In [3]: l1 = [10, 20, 30]
        l1
```

```
Out[3]: [10, 20, 30]
```

```
In [4]: len(l1)
```

```
Out[4]: 3
```

```
In [5]: l1
```

```
Out[5]: [10, 20, 30]
```

```
In [6]: l1.append(40)
        l1.append(50)
        l1.append(60)
        l1.append(10)
```

```
In [7]: l1
```

```
Out[7]: [10, 20, 30, 40, 50, 60, 10]
```

```
In [8]: len(l1)
```

```
Out[8]: 7
```

In [9]: `l1.append(70, 80)`

```
-----
-----
TypeError                                Traceback (most recent
call last)
Cell In[9], line 1
----> 1 l1.append(70, 80)

TypeError: list.append() takes exactly one argument (2 given)
```

In [10]: `l1`

Out[10]: `[10, 20, 30, 40, 50, 60, 10]`

In [11]: `l1.append(['a', 'b', 'c'])`

In [12]: `l1`

Out[12]: `[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]`

In [14]: `l2 = [34, 5.3, 'jyothi', 1+2j, True]`
`l2`

Out[14]: `[34, 5.3, 'jyothi', (1+2j), True]`

In [15]: `print(l)`
`print(l1)`
`print(l2)`

```
[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'jyothi', (1+2j), True]
```

In [16]: `print(len(l))`
`print(len(l1))`
`print(len(l2))`

```
0
8
5
```

```
In [17]: print(id(l))  
         print(id(l1))  
         print(id(l2))
```

```
1259783118592  
1259808159168  
1259814340608
```

```
In [18]: print(type(l))  
         print(type(l1))  
         print(type(l2))
```

```
<class 'list'>  
<class 'list'>  
<class 'list'>
```

```
In [19]: l1
```

```
Out[19]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [20]: ['a', 'b', 'c'] in l1
```

```
Out[20]: True
```

```
In [21]: l1
```

```
Out[21]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [22]: 100 in l1
```

```
Out[22]: False
```

```
In [23]: l2
```

```
Out[23]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In [24]: l3 = l2.copy()
```

```
In [25]: l3
```

```
Out[25]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In [26]: print(id(l2))
         print(id(l3))
```

```
1259814340608
1259814373184
```

```
In [27]: print(l)
         print(l1)
         print(l2)
         print(l3)
```

```
[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'jyothi', (1+2j), True]
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [28]: l3
```

```
Out[28]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In [29]: len(l3)
```

```
Out[29]: 5
```

```
In [30]: l3.clear()
```

```
In [31]: l3
```

```
Out[31]: []
```

```
In [32]: len(l3)
```

```
Out[32]: 0
```

```
In [33]: del l3
```

```
In [34]: print(l)
         print(l1)
         print(l2)
```

```
[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [35]: 12.count(5.3)
```

```
Out[35]: 1
```

```
In [36]: 11.count(10)
```

```
Out[36]: 2
```

```
In [37]: 11
```

```
Out[37]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [38]: 11
```

```
Out[38]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [39]: 11[:]
```

```
Out[39]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [40]: 11[-1]
```

```
Out[40]: ['a', 'b', 'c']
```

```
In [41]: 11
```

```
Out[41]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [42]: 11[100]
```

```
-----  
-----  
IndexError                                Traceback (most recent  
call last)  
Cell In[42], line 1  
----> 1 11[100]  
  
IndexError: list index out of range
```

In [43]: 11

Out[43]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [44]: 11[4:]

Out[44]: [50, 60, 10, ['a', 'b', 'c']]

In [45]: 11

Out[45]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [46]: 11[3:-1]

Out[46]: [40, 50, 60, 10]

In [47]: 11

Out[47]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [48]: 11[0:10:3]

Out[48]: [10, 40, 10]

In [49]: 11

Out[49]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [50]: 11[:,1]

Out[50]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [51]: 11[:,2]

Out[51]: [10, 30, 50, 10]

In [52]: 11

Out[52]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [53]: `l1.count(10)`

Out[53]: 2

In [54]: `l1`

Out[54]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [55]: `l1[::-2]`

Out[55]: [['a', 'b', 'c'], 60, 40, 20]

In [56]: `l1`

Out[56]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [57]: `l1[::-3]`

Out[57]: [['a', 'b', 'c'], 50, 20]

In [58]: `l1`

Out[58]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

In [59]: `l1[4:-3]`

Out[59]: [50]

In [60]: `print(l)
print(l1)
print(l2)`

```
[]  
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]  
[34, 5.3, 'jyothi', (1+2j), True]
```

In [61]: `l1`

Out[61]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [62]: l1[0]
```

```
Out[62]: 10
```

```
In [63]: l1[0] = 100
```

```
In [64]: l1
```

```
Out[64]: [100, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [65]: l1[-1] = 'jyothi'
l1
```

```
Out[65]: [100, 20, 30, 40, 50, 60, 10, 'jyothi']
```

```
In [66]: l1
```

```
Out[66]: [100, 20, 30, 40, 50, 60, 10, 'jyothi']
```

```
In [67]: print(l)
print(l1)
print(l2)
```

```
[]
[100, 20, 30, 40, 50, 60, 10, 'jyothi']
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [68]: l.extend(l2)
```

```
In [69]: print(l)
print(l1)
print(l2)
```

```
[34, 5.3, 'jyothi', (1+2j), True]
[100, 20, 30, 40, 50, 60, 10, 'jyothi']
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [72]: l.extend(l1)
1
```

```
Out[72]: [34,
          5.3,
          'jyothi',
          (1+2j),
          True,
          100,
          20,
          30,
          40,
          50,
          60,
          10,
          'jyothi',
          100,
          20,
          30,
          40,
          50,
          60,
          10,
          'jyothi']
```

```
In [73]: l.index('jyothi')
```

```
Out[73]: 2
```

```
In [74]: l1
```

```
Out[74]: [100, 20, 30, 40, 50, 60, 10, 'jyothi']
```

```
In [75]: l1.append(10)
```

```
In [76]: l1
```

```
Out[76]: [100, 20, 30, 40, 50, 60, 10, 'jyothi', 10]
```

```
In [77]: l1.index(10)
```

```
Out[77]: 6
```

In [78]: l1

Out[78]: [100, 20, 30, 40, 50, 60, 10, 'jyothi', 10]

In [79]: l1.insert(2, 25)

In [80]: l1

Out[80]: [100, 20, 25, 30, 40, 50, 60, 10, 'jyothi', 10]

In [86]: l1.pop()

Out[86]: 'jyothi'

l1.pop()

In [82]: l1

Out[82]: [100, 20, 25, 30, 40, 50, 60, 10, 'jyothi']

In [83]: l

Out[83]: [34,
5.3,
'jyothi',
(1+2j),
True,
100,
20,
30,
40,
50,
60,
10,
'jyothi',
100,
20,
30,
40,
50,
60,
10,
'jyothi']

In [87]: `l.pop()`

Out[87]: 10

In [89]: `l.pop(2)`

Out[89]: 'jyothi'

In [91]: `l[3]`

Out[91]: True

In [92]: `l.remove(1+2j)`

In [93]: `l`

Out[93]: [34, 5.3, True, 100, 20, 30, 40, 50, 60, 10, 'jyothi', 100, 20, 30, 40, 50]

In [94]: `l.reverse()`

In [95]: `l`

Out[95]: [50, 40, 30, 20, 100, 'jyothi', 10, 60, 50, 40, 30, 20, 100, True, 5.3, 34]

In [96]: `print(l)`
`print(l1)`
`print(l2)`

[50, 40, 30, 20, 100, 'jyothi', 10, 60, 50, 40, 30, 20, 100, True, 5.3, 34]

[100, 20, 25, 30, 40, 50, 60, 10]

[34, 5.3, 'jyothi', (1+2j), True]

In [97]: 12.sort()

```
-----  
-----  
TypeError                                Traceback (most recent  
call last)  
Cell In[97], line 1  
----> 1 12.sort()  
  
TypeError: '<' not supported between instances of 'str' and  
'float'
```

In [98]: 1

Out[98]: [50, 40, 30, 20, 100, 'jyothi', 10, 60, 50, 40, 30, 20, 100, True,
5.3, 34]

In [99]: 11.sort()

In
[100]: 11

Out[100]: [10, 20, 25, 30, 40, 50, 60, 100]

In
[101]: 11.sort(reverse= True)
11

Out[101]: [100, 60, 50, 40, 30, 25, 20, 10]

In
[102]: 12

Out[102]: [34, 5.3, 'jyothi', (1+2j), True]

In
[103]: 13 = [3 , 1, 2]
13

Out[103]: [3, 1, 2]

In
[104]: 13.sort() # paramteter tunning
13

Out[104]: [1, 2, 3]

```
In      13.sort(reverse=True)
[105]:  13
```

```
Out[105]: [3, 2, 1]
```

```
In      11
[106]:
```

```
Out[106]: [100, 60, 50, 40, 30, 25, 20, 10]
```

```
In      12
[107]:
```

```
Out[107]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In      12[2]
[108]:
```

```
Out[108]: 'jyothi'
```

```
In      12
[109]:
```

```
Out[109]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In      12[2][0]
[110]:
```

```
Out[110]: 'j'
```

```
In      print(12[2][0])
[111]:  print(12[2][1])
        print(12[2][2])
        print(12[2][3])
        print(12[2][4])
        print(12[2][5])
```

```
j
y
o
t
h
i
```

```
In      1
[112]:
```

```
Out[112]: [50, 40, 30, 20, 100, 'jyothi', 10, 60, 50, 40, 30, 20, 100, True,
           5.3, 34]
```

```
In      for i in l:  
[113]:     print(i)
```

```
50  
40  
30  
20  
100  
jyothi  
10  
60  
50  
40  
30  
20  
100  
True  
5.3  
34
```

```
In      for i in enumerate(l):  
[114]:     print(i)
```

```
(0, 50)  
(1, 40)  
(2, 30)  
(3, 20)  
(4, 100)  
(5, 'jyothi')  
(6, 10)  
(7, 60)  
(8, 50)  
(9, 40)  
(10, 30)  
(11, 20)  
(12, 100)  
(13, True)  
(14, 5.3)  
(15, 34)
```

```
In      11  
[115]:
```

```
Out[115]: [100, 60, 50, 40, 30, 25, 20, 10]
```

```
In      all(11)  
[116]:
```

```
Out[116]: True
```

```
In any(l1)
[117]:
```

```
Out[117]: True
```

```
In l1.append(0)
[118]:
```

```
In l1
[119]:
```

```
Out[119]: [100, 60, 50, 40, 30, 25, 20, 10, 0]
```

```
In all(l1)
[120]:
```

```
Out[120]: False
```

```
In any(l1)
[121]:
```

```
Out[121]: True
```

```
In any(l1)
[127]:
```

```
Out[127]: True
```

```
In list1 =[]
[130]:
```

```
In print(type(list1))
[131]:
<class 'list'>
```

```
In list2
[134]:
```

```
Out[134]: [10, 30, 60]
```

```
In list3
[135]:
```

```
Out[135]: [10.77, 30.66, 60.89]
```

```
In L1 = [1,2,3,4,0]
[138]:
```

```
In      all(L1)
[139]:
```

```
Out[139]: False
```

```
In      any(L1)
[140]:
```

```
Out[140]: True
```

```
In      L2 = [1,2,3,4,True,False]
[141]:
```

```
In      all(L2)
[142]:
```

```
Out[142]: False
```

```
In      any(L2)
[143]:
```

```
Out[143]: True
```

```
In      L3 = [1,2,3,True]
[144]:
```

```
In      all(L3)
[145]:
```

```
Out[145]: True
```